

جامعة أم القرى
UMM AL-QURA UNIVERSITY

Course: Software Engineering Fundamentals

Project Title: Simple Budget Planner
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Section:14

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part2

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Functional Requirements

1- The system allows the user to add income from different sources.
2- The system allows the user to add and record daily expenses.
3- The system allows the user to view all transactions (income and expenses).
4- The system calculates and displays the remaining balance automatically.
5- The system allows the user to set a monthly budget.
6- The system sends alerts when the user is close to exceeding the budget.

Non-Functional Requirements

1- The system should be easy to use for all users, especially beginners with limited financial knowledge.
2- The system should respond quickly when users add or view their financial data.
3- The system should ensure the security and privacy of user data through basic authentication.
4- The system should be available and accessible on different devices such as smartphones and computers.
5- The system should provide clear and simple user interfaces for better user experience.

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Target Users & personas
To understand how the system will work with the people who use it we made up two examples of users.

persona 1
* Name: Sara Al-Harbi
* Age: 21
* Job: University Student
* Story: Sara Al-Harbi is a student at Umm Al-Qura University. Her family gives her some money every month. She has a hard time using it wisely and usually runs out of money before the month is over.
* What Sara Al-Harbi wants to do:
. Keep track of what she spends every day
+ Not spend too much money
+ Save some of her allowance
* What Sara Al-Harbi needs:
. A simple way to keep track of her spending
+ A reminder when she is about to spend much money
* What bothers Sara Al-Harbi:
. She does not know where her money is going
. She uses a messy system to keep track of her spending

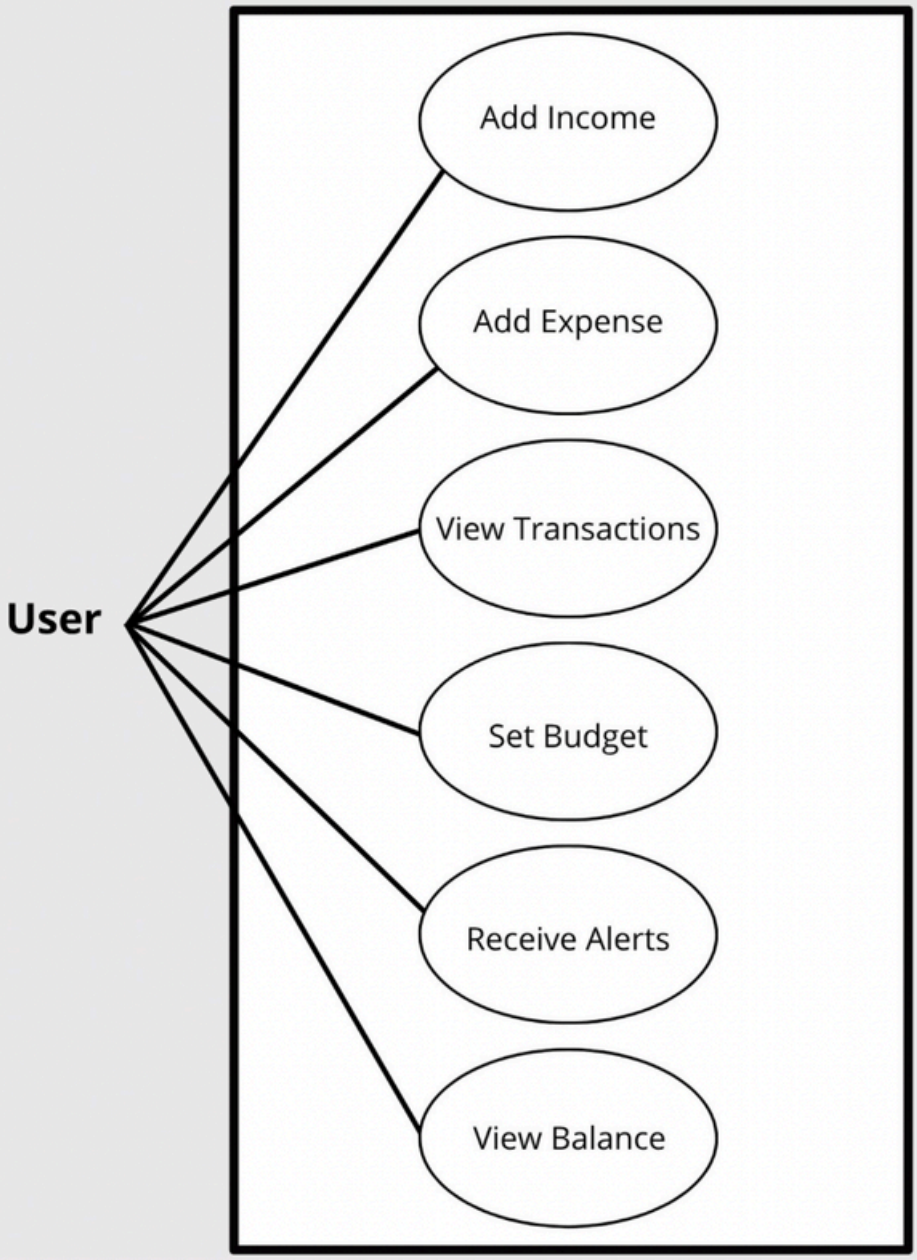
Persona 2

* Name: Abdulrahman Al-Qahtani
* Age: 26
* Job: Junior Employee
* Story: Abdulrahman Al-Qahtani has a job. He gets paid every month. He wants to use his money to pay his bills on time and save some. He does not know how to handle his money well because he is new at this.
* What Abdulrahman Al-Qahtani wants to do:
. Manage his money
+ Pay his bills on time
. Save money
* What Abdulrahman Al-Qahtani needs:
. A clear picture of his money
+ A report every month
* What bothers Abdulrahman Al-Qahtani:
. He has a hard time managing his spending
+ He is not good at being careful, with his money

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Simple Budget Planner

This diagram shows how the user interacts with the Simple Budget Planner system. The user can perform actions such as adding income, recording expenses, viewing transactions setting a budget, receiving alerts, and checking the balance



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Use Case 1: Add Income

Brief Description
The user enters the income amount and its source, and the system saves it and updates the balance automatically.

Goal
To enable the user to record any amount received from a salary, monthly allowance, or any other source.

Actor
User

Basic Steps
1- The user opens the “Add Income” page.
2- The user enters the income amount and its source (e.g., salary, monthly allowance, freelance work).
3- The user clicks the “Save” button.
4- The system saves the income in the database.
5- The system automatically updates the current balance.
6- The system displays a confirmation message indicating that the process was successful.

Outcome
The income is successfully recorded and the balance is updated.

Use Case 2: Add Expense

Brief Description
The user records a daily expense, and the system deducts the amount from the balance and compares it with the monthly budget.

Goal
To help the user track daily expenses and understand their impact on the monthly budget.

Actor
User

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Use Case 2: Add Expense

Basic Steps

1- The user opens the “Add Expense” page.
2- The user enters the expense amount and its category (e.g., food, transportation, shopping).
3- The user clicks the “Save” button.
4- The system deducts the amount from the balance.
5- The system checks the monthly budget:
If the user is close to exceeding the budget, the system sends an alert.
6- The system displays a confirmation message indicating that the expense has been recorded.

Outcome
The expense is recorded, the balance is updated, and an alert is sent if necessary.

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Student	Task Completed
Ghadi Bandar Mukhtar	Problem Statement Proposed Software Solution Use Case Descriptions
Yara Shadi Al Ansari	Non Functional Requirements
Ghala Hussein Al Sharif	Target users and Personas
Joud Abdullah Al osaime	Functional Requirements
Anhar mohammed al nafiei	Created the Use Case Diagram and explained user interactions



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